John topic 1: Major/Interest/Profession VS. AI Usage/Concern

Question: (The quote will explain the purpose of the question, since some questions are not really straightforward)

1. What is your current academic major? (Distinguishing between humanities/sciences)

Type: Categorical (Nominal)

Answers: Computer Science, Computer Engineering, Data Science, Biology, Business, Psychology, Mathematics, Arts/Design, Undeclared, Other

2. Which field best describes your primary career interest? (Distinguishing between specialized and everyday usage)

Type: Categorical (Nominal)

Answers: Software Development, AI/Machine Learning, Data Science/Analytics, Cybersecurity, Hardware/Engineering, Research/Science, Business/Finance, Healthcare, Creative/Design, Education, Other, Undecided

3. How many days per week do you typically use AI tools?

Type: Numerical (Discrete)

Answers: 0–7 days

4. On a typical day when you use AI, approximately how many minutes do you actively spend using it?

Type: Numerical (Continuous)

Answers: Nonnegative number of minutes; not applicable if you never use AI

5. For which purposes do you use AI? Select all that apply.(Distinguishing between specialized and everyday usage)

Type: Categorical (Multiple Response)

Answers: Homework/Studying, Coding, Research, Writing, Brainstorming, Creative Projects, Personal Advice, Entertainment, Other, I do not use AI

6. How knowledgeable do you consider yourself about AI?

Type: Ordinal (0–5)

Answers: 0 = No knowledge, 5 = Very knowledgeable

7. How much do you trust AI-generated information?

Type: Ordinal (0–5)

Answers: 0 = No trust, 5 = Complete trust; Not sure/no experience

8. How concerned are you about AI reducing job opportunities in your intended career field?

Type: Ordinal (0–5)

Answers: 0 = Not concerned, 5 = Extremely concerned; Undecided about career

9. How capable do you think AI will be of performing the main tasks in your intended career within the next 10 years?

Type: Ordinal (0–5)

Answers: 0 = Not capable, 5 = Fully capable; Undecided about career

10. What percentage of your academic work do you estimate involves using AI assistance?

Type: Numerical (0–100%)

Answers: 0–100%; approximate percentage

11. What is your biggest concern about AI?

Type: Categorical (Nominal)

Answers: Job Replacement, Misinformation, Privacy, Bias/Unfairness, Overdependence, Academic Integrity, Environmental Impact, Other, No Major Concern

Can be represented via a graph constructed based on the problem:

1. Histogram, Q4: Distribution of daily time spent using artificial intelligence.

2. Box plot, Q2, Q4: Which professional interest groups spend more time using artificial intelligence?

3. Scatter plot, Q3, Q8: Is frequent use of artificial intelligence associated with concerns regarding career development?

4. Heatmap, Q1, Q5: For what different purposes do professionals in various fields use artificial intelligence?

5. Grouped bar chart, Q2, Q8: How do levels of career-related concern differ among practitioners across various interest areas?

6. Radar chart, Q2, Q6–Q9: How do individuals with different professional interests vary in their knowledge, trust, level of concern, and perceived capabilities regarding artificial intelligence?

7. Parallel coordinates, Q3, Q6–Q9: What combinations of artificial intelligence usage, knowledge, trust, and concerns emerge from the data?

8. Scatter plot matrix, Q3, Q4, Q6–Q9: What relationships exist among variables related to artificial intelligence?

3 Testable Statistical Hypotheses:

Hypothesis 1: The Relationship Between Frequency of AI Use and Career-Related Concerns

Students who use artificial intelligence more frequently tend to differ in their level of concern regarding whether AI might reduce future employment opportunities.

Variables: Q3 and Q8

Statistical Test: Spearman's Rank Correlation Coefficient

Visualization: Scatter plot with a trend line

Hypothesis 2: Career Interests and Primary Concerns Regarding AI

The distribution of primary concerns about artificial intelligence varies among students with different career interests.

Variables: Q2 and Q11

Statistical Test: Chi-square Test of Independence

Visualization: Heatmap or 100% stacked bar chart

Hypothesis 3: Perceived AI Capabilities and Concerns About Career Development

Students who expect artificial intelligence to take on a greater number of tasks in their future careers tend to be more concerned about their employment prospects.

Variables: Q9 and Q8

Statistical Test: Spearman's Rank Correlation Coefficient

Visualization: Bubble chart or jittered scatter plot

John topic 2: Combination of AI Usage and Student's Academic/Profession Interest

Main question: How do students' academic and professional interests relate to their AI-related habits, their level of reliance on AI, and their ability to solve problems independently? (Does more frequent use of artificial intelligence lead students to become increasingly dependent on it? And does this dependency vary based on their academic or professional interests?)

Note for this question – Possible answer for this question, for example:

Students who feel interested in coding are more likely to rely on AI compared with students who feel interested in cyber security?

Student who use AI often will find it harder to accomplish questions independently?

Survey question: (Q1-3 from topic 1, since those are some general questions in this topic)

1. What is your current academic major? (Distinguishing between humanities/sciences)

Type: Categorical (Nominal)

Answers: Computer Science, Computer Engineering, Data Science, Biology, Business, Psychology, Mathematics, Arts/Design, Undeclared, Other

2. How many days per week do you typically use AI tools?

Type: Numerical (Discrete)

Answers: 0–7 days

3. On a typical day when you use AI, approximately how many minutes do you actively spend using it?

Type: Numerical (Continuous)

Answers: Nonnegative number of minutes; not applicable if you never use AI

4. For what purpose do you most frequently use artificial intelligence in your coursework? (possible answer: generating complete solutions, obtaining hints, seeking explanations, or creating practice exercises; select "do not use AI" if applicable) Categorical

5. When you encounter a difficult assignment problem, how many minutes do you typically spend attempting it on your own before turning to artificial intelligence? Numerical

6. To what extent do you rely on artificial intelligence to complete your academic assignments? (Scale: 0–5) Ordinal

7. How confident are you in your ability to solve academic problems without using artificial intelligence? (Scale: 0–5) Ordinal

8. When artificial intelligence provides a solution, to what extent are you able to explain that solution on your own? (Scale: 0–5) Ordinal

9. How does using artificial intelligence affect your ability to solve problems independently? (Enhances, No impact, Diminishes, Unsure) Categorical

10. What are your typical exam scores in courses related to your major? (0–100%)Numerical

TOPIC 3: Homework vs. exam gap

Broad question: Do students who rely on AI for homework do worse when it isn't allowed?

Main question: How does the gap between students' homework and exam scores depend on how much they use AI, what they use it for, and how soon they turn to it?

Questions:

1. What do you mostly use AI for in your classes?

   Options: (Getting full solutions or working code / Getting hints when stuck / Explaining concepts or code / Making practice problems or mock exams / I don't use AI for schoolwork) multiple choice

2. In a typical week during the quarter, about how many hours do you use AI for schoolwork?

   Options: (number)

3. When you're stuck, how long do you usually keep trying before asking AI?

   Options: (Right away / Under 15 minutes / 15-60 minutes / Over an hour / I don't ask AI when stuck) multiple choice

4. What's your typical homework score in your CS classes?

   Options: (number, 0-100)

5. What's your typical exam score in your CS classes?

   Options: (number, 0-100)

TOPIC 4: Why some students use AI more than others

Broad question: Is heavy AI use about pressure, the easy way out, or what everyone else does, and what holds some students back?

Main question: How does how much students use AI depend on their obligations outside school, what they do when stuck, what holds them back, and how much they think classmates use it?

Questions:

1. In a typical week during the quarter, about how many hours do you use AI for schoolwork?

   Options: (number)

2. In a typical week during the quarter, about how many hours go to obligations outside school (paid work, family responsibilities, other commitments)?

   Options: (number)

3. When homework is due soon and you're stuck, what do you usually do?

   Options: (Keep trying on my own / Ask a classmate or TA / Ask AI for a hint / Ask AI for the answer / Turn in what I have) multiple choice

4. What mainly keeps you from using AI more than you do?

   Options: (I want to learn it myself / I don't trust its answers / Class rules / It feels like cheating / Nothing, I use it as much as I want) multiple choice

5. About how many hours a week do you think a typical classmate uses AI for schoolwork?

   Options: (number)

TOPIC 5: What separates top students from everyone else

Broad question: Do top students work harder, or work differently?

Main question: How do students' GPAs depend on how much they study, how they study, how often they go to class, and how well they can predict their own exam scores?

Questions:

1. What is your current college GPA?

   Options: (number, 0.0-4.0; optional, skip if you don't have one yet)

2. In a typical week during the quarter, about how many hours do you study outside of class?

   Options: (number)

3. About what percent of your lectures and discussion sections do you usually attend?

   Options: (number, 0-100)

4. What's your main way of studying for exams?

   Options: (Re-reading notes, slides, or lecture videos / Doing practice problems or past exams / Quizzing myself with flashcards / Writing summaries or study guides / Studying with a group) multiple choice

5. When you get an exam back, how does your score usually compare to what you expected?

   Options: (Usually lower than I expected / Usually about what I expected / Usually higher than I expected / I don't usually have an expectation) multiple choice

TOPIC 6(Jovanni): How much do you/should you trust AI

Broad Question: Does knowing the info was AI generated affect how you see it and trust it

Questions:

1. How trustworthy do you find AI on these topics

   options: Matrix with 1-5, 1 being not trustworthy at all, 5 being very trustworthy
2. Would you share info you have gathered with AI to another person

   options: Yes, No, Maybe (def change this)
3. How often do you fact check AI info gathered by you or another

   options: 1-5, 1 being not very often, 5 being very often
4. If you found out later that your info was gathered through AI would you lose trust in it

   options:1-5 1 being lose all trust, 5 being same trust

TOPIC 7(Jovanni): Ai in healthcare

broad question: How does AI in healthcare such as AI generated summaries compared to news from a healthcare professional affect perceived trustworthiness

1. How trustworthy would you find this healthcare advice (give example of real healthcare advice)

   options: 1-5 rating of how trustworthy they find it

2. How accurate do you believe this healthcare advice to be (give AI generated example of healthcare advice)

   options: 1-5 rating of how trustworthy they find it

3. How often would you use AI to ask health related questions

   options: 1-5 rating how often

4. How uncomfortable would you be to learn your health care advice was from AI

   options: 1-5 how uncomfortable to comfortable

Topic 8 (Khant): How does your personality affect your AI usage and concern

Broad Question: Can a person's personality or preferences affect how much AI they use or the concern they have about AI?

Main Question: Do specific personality traits lead to a person using more or less AI and/or how concerned they are about future development in AI fields.

Example questions might be targeted more towards figuring out a person's personality then their usage of AI:

1. Would you consider yourself an introvert or extrovert?
2. Do you tend to finish assignments as they are assigned or just before they are due?
3. When trying to solve a social problem, do you try to take the more logical approach or consider other's feelings first?
4. How often do you think you use AI when completing an academic related task or assignment?
5. Do you use AI in your daily life for things unrelated to work or school? (Qualitative question follow up could be something like "what do you usually use AI for in these cases?")
6. Considering the recent developments in AI technology both in search algorithms and more integrated use of AI, do you think it will have a positive or negative effect on future fields?

Topic 9 (Shiyao): AI Agent Usage Patterns and Preferences Among College Students

Q1: Have you ever used an AI agent?
Ans: Yes/No

Q2: How often do you use AI agents?
Ans: Never, Monthly, Weekly, Daily

Q3: Which AI agents have you used?
Ans: Codex, Claude, Other, I don't use AI agent

Q4: Which AI agent do you use most frequently?

Q5: What do you use AI agents for?(Select all that apply**)**

Ans: Homework and studying

Programming and debugging

Research and information gathering

Writing and editing

Email and schedule management

Everyday personal tasks

Other

Q6: What is your primary purpose for using AI agents?

Q7: How much do you trust AI agents to provide accurate results?

Q8: How often do you verify the results produced by AI agents?

Q9: How much autonomy are you willing to give an AI agent?

Q10: Which tasks would you allow an AI agent to complete without your approval?

Q11: How do you typically use AI tools? (Which of the following best describes your most common way of using AI?)

I mainly use AI to answer questions or generate content.

AI helps me prepare tasks, but I perform the actions myself.

I allow AI agents to perform some tasks or actions on my behalf.

I do not use AI tools.

Q12: Have you ever allowed an AI agent to take actions on your behalf?

No, I have never allowed AI to perform actions.

Yes, but I review and approve actions before execution.

Yes, I allow AI to complete certain tasks without approving every action.

I'm not sure.